

```

from machine import Pin, ADC
from time import sleep_ms

# =====
# SENSORS
# =====

# Change GPIO numbers if needed
light_sensor = ADC(Pin(1))
moisture_sensor = ADC(Pin(2))

# Sensor thresholds
LIGHT_THRESHOLD = 120
MOISTURE_THRESHOLD = 4000

# =====
# STEPPER MOTOR (28BYJ-48)
# =====

IN1 = Pin(5, Pin.OUT)
IN2 = Pin(6, Pin.OUT)
IN3 = Pin(7, Pin.OUT)
IN4 = Pin(8, Pin.OUT)

motor_pins = [IN1, IN2, IN3, IN4]

# Half-step sequence
sequence = [
    [1,0,0,0],
    [1,1,0,0],
    [0,1,0,0],
    [0,1,1,0],
    [0,0,1,0],
    [0,0,1,1],
    [0,0,0,1],
    [1,0,0,1]
]

# About 90 degrees
TARGET_STEPS = 512

current_position = 0

# =====
# MOTOR FUNCTIONS
# =====

def release_motor():
    for pin in motor_pins:
        pin.value(0)

def step_motor(steps):
    global current_position

```

```

direction = 1 if steps > 0 else -1

for _ in range(abs(steps)):
    current_position += direction

    pattern = sequence[current_position % 8]

    for pin, value in zip(motor_pins, pattern):
        pin.value(value)

    sleep_ms(2)

release_motor()

def move_to(target_position):
    global current_position

    difference = target_position - current_position

    if difference != 0:
        step_motor(difference)

# =====
# MAIN LOOP
# =====

while True:

    moisture_value = moisture_sensor.read()
    light_value = light_sensor.read()

    print(
        "Moisture:", moisture_value,
        " Light:", light_value
    )

    moisture_detected = moisture_value < MOISTURE_THRESHOLD
    dark_detected = light_value < LIGHT_THRESHOLD

    if moisture_detected or dark_detected:

        if current_position != TARGET_STEPS:
            print("TRIGGERED - MOVING TO 90 DEGREES")
            move_to(TARGET_STEPS)

    else:

        if current_position != 0:
            print("NO TRIGGER - RETURNING TO 0 DEGREES")
            move_to(0)

    sleep_ms(500)

```